

Varun Shelke

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EDUCATION

University of Pittsburgh

Master of Science in Information Science | GPA: 3.917

Pittsburgh, PA

Aug. 2024 – Apr 2026

University of Pune

Bachelor of Engineering in Computer Engineering

Pune, India

Aug. 2016 – Jul 2020

EXPERIENCE

AI Engineer

Pitt AWS Cloud Innovation Center

May 2025 – Present

Pittsburgh, PA

- Built a **production GenAI pipeline** on Amazon Bedrock with OCR and human-in-the-loop validation, eliminating **10+ hrs/week** of manual entry across **20+ daily image inputs**.
- Deployed an **enterprise AI analytics system** using **Python**, **Pydantic-based** tool-use, and **vector embeddings**, enabling semantic search across **1M+ unstructured records** for real-time client insights.
- Built an **event-driven** GenAI solution on **AWS Lambda**, **SNS**, and **SQS** (TypeScript/Node.js), using **LLM intent classification** to automate **67% of 100K+ customer interactions** with human-in-the-loop routing.
- Built reusable **infrastructure-as-code** assets via **AWS CDK** for **3 production GenAI apps**, with **CI/CD** pipelines integrating **security scanning**, **linting**, and **GitOps workflows**.

Engineer

Nagarro

May 2022 – Aug. 2024

Pune, India

- **Led** a 3-engineer pod building a **high-throughput** Order Management System in **Java/Spring Boot** with SAP Commerce via APIs, **reducing cancellations by 85%** and **scaling inventory to 10x** across 62 centers.
- **Architected** a **fulfillment routing engine** in Java, algorithmically **balancing inventory**, lead times, and **cost** across 62 centers to **eliminate overselling** at 2K-5K daily order peaks.
- Built **reliable**, well-documented order infrastructure with **CronJobs** and **retry mechanisms**, improving fault tolerance and reducing manual intervention across distributed operations.

Programmer Analyst

Cognizant Consulting

Nov. 2020 – May 2022

Pune, India

- Delivered 20+ production features for **Fortune 500 clients** in Java/Spring Boot and SAP Commerce Cloud, including real-time inventory **API integrations** with SAP ECC.
- Increased **unit test coverage** from **72% to 89%** by implementing comprehensive **JUnit/Mockito** test suites and resolving **SonarQube** code quality issues, improving codebase maintainability for 15-engineer team.
- Engineered **bi-directional** integration between e-commerce systems and **Salesforce CRM** ensuring accurate order and customer data synchronization across enterprise platforms.

ACCOMPLISHMENTS & LEADERSHIP

- Built [diving analytics platform](#) deployed by **Pitt Diving** and garnered interest from **USA Diving** team
- **1st place** at [KPMG Cup Showcase](#) for strategic analysis projecting **\$50M** value creation for PPG Industries
- **2nd place** in [Tartanhacks Hackathon](#) at **Carnegie Mellon University** (279 teams).
- Mentored **2 interns** at AWS Cloud Innovation Center, guiding development of production GenAI applications
- Received **Engineering Excellence** award at Nagarro for outstanding technical contributions and project delivery

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, Typescript, SQL, GraphQL, MySQL

Frameworks: Spring Boot, FastAPI, Node.js, React.js, SAP Commerce Cloud, Fluent Commerce

Cloud & Tools: AWS (CDK, Lambda, S3, Bedrock, SNS, SQS, DynamoDB, End User Messaging), Pydantic, LangChain, LangGraph, Redis, Docker, Kubernetes, Terraform, Git, Jira, Postman, Claude Code, Kiro CLI.